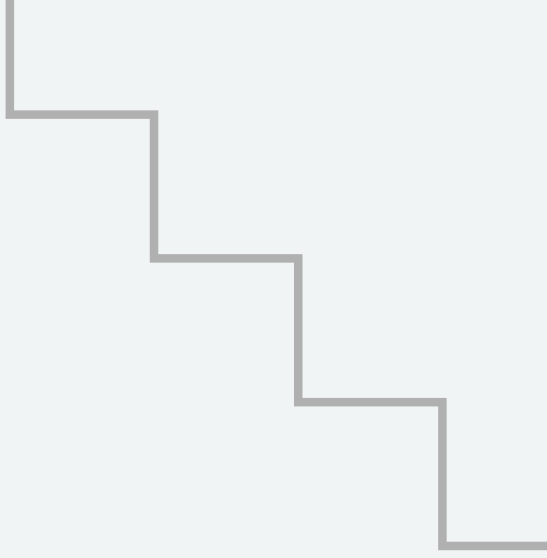


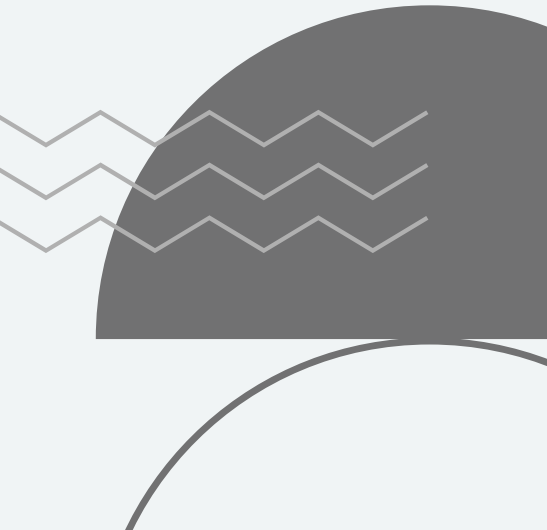
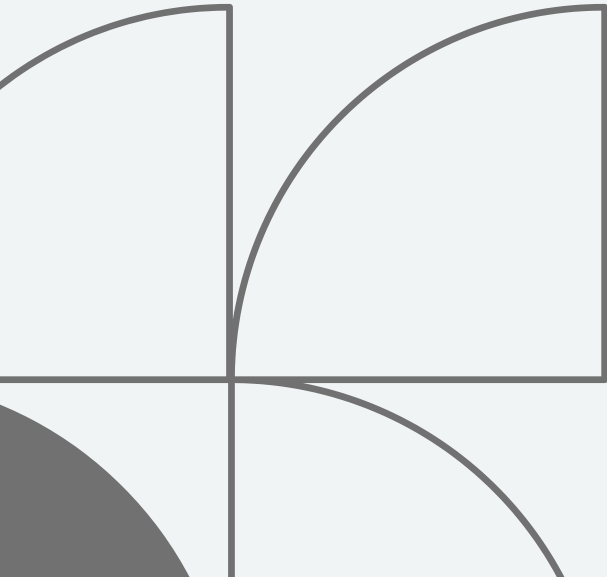


Object-Oriented Programming (Java)

PROJECT PRESENTATION

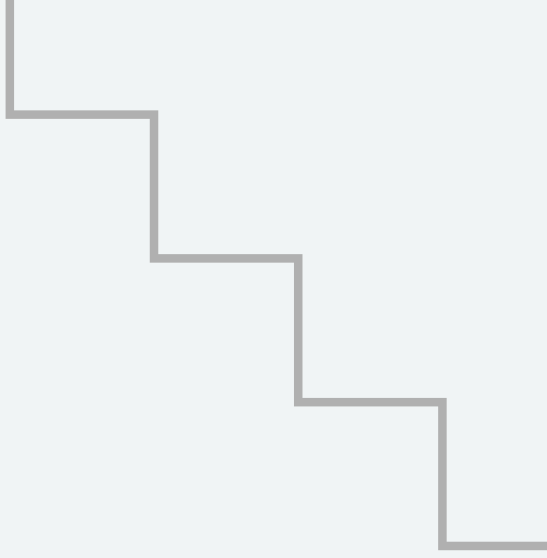
SMART WAREHOUSE GRID LOCATOR

By Ravi Sharma





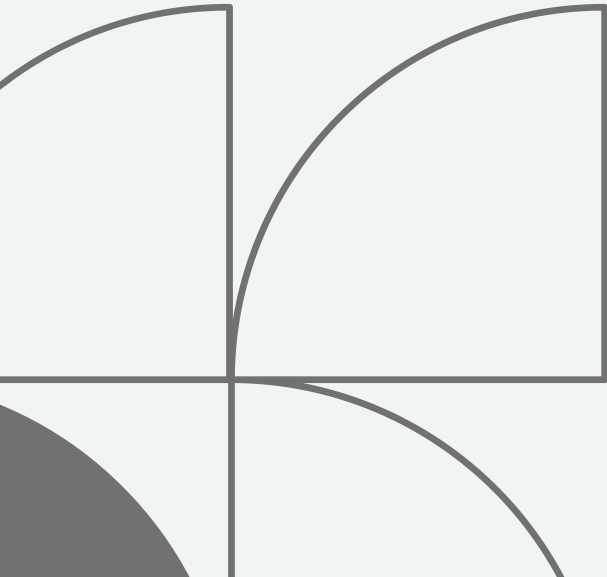
INTRODUCTION



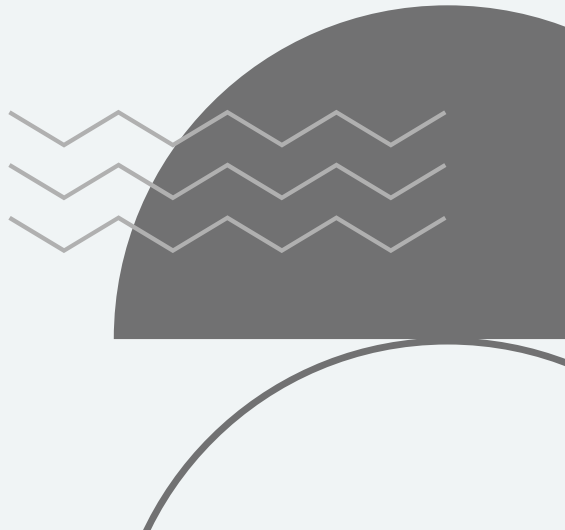
Modern warehouses store thousands of products and require efficient item tracking systems.

The Smart Warehouse Grid Locator is a Java-based application that simulates a warehouse using a 2D grid structure.

OBJECTIVE

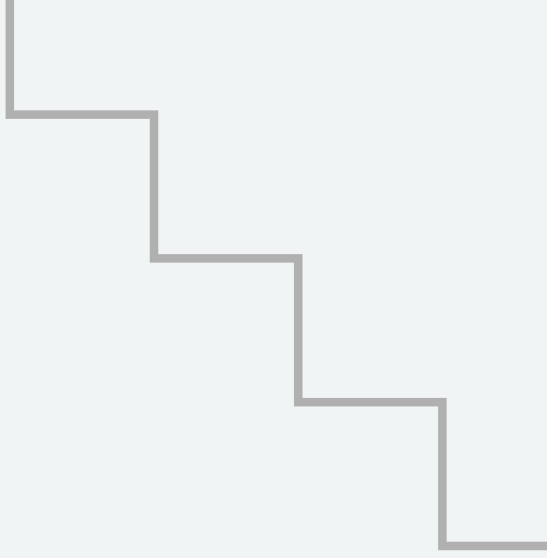
- Store items in a warehouse grid.
 - Track item locations using coordinates.
 - Search items quickly using Item ID.
 - Demonstrate Object-Oriented Programming concepts.
- 

Technology Used

- Java
 - OOP Concepts
 - 2D Arrays
 - HTML, CSS, JavaScript (Frontend)
- 



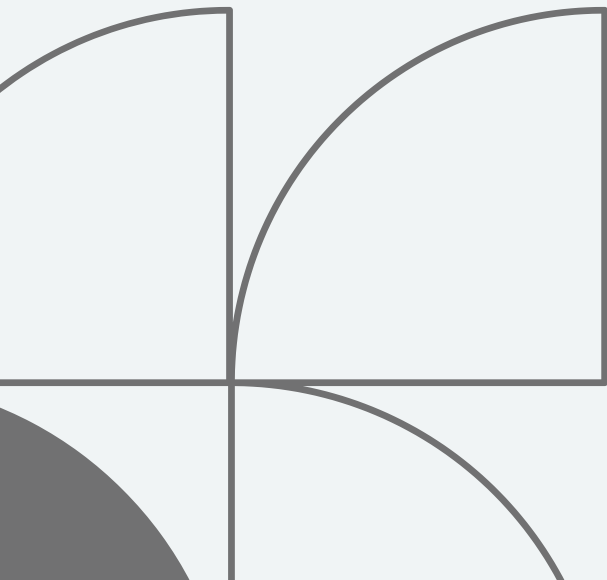
FEATURES



ITEM MANAGEMENT

- Add new items
- Update item quantity
- Remove items

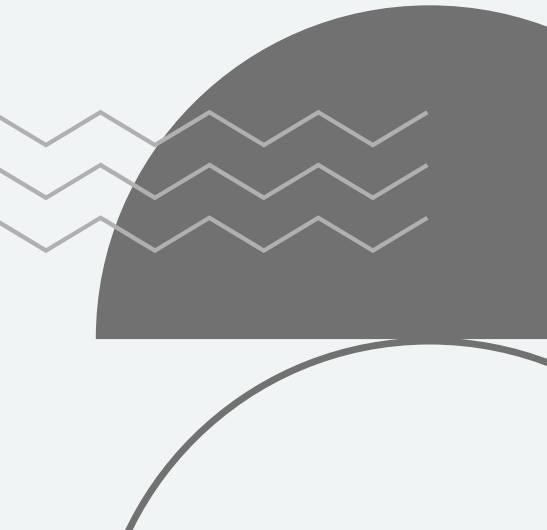
WAREHOUSE VISUALIZATION

- Grid-based warehouse layout
 - Occupied and empty slot identification
- 

SEARCH FUNCTIONALITY

- Search by Item ID
- Search by Item Name
- Display exact row and column location

OOP IMPLEMENTATION

- Encapsulation
 - Classes and Objects
 - Constructors
 - Method-based operations
- 

WORKING OF THE PROJECT

HOW THE SYSTEM WORKS

Step 1: Initialize Warehouse

- Create a 2D grid (e.g., 5x5)

Step 2: Add Items

- Store item information in specific grid positions

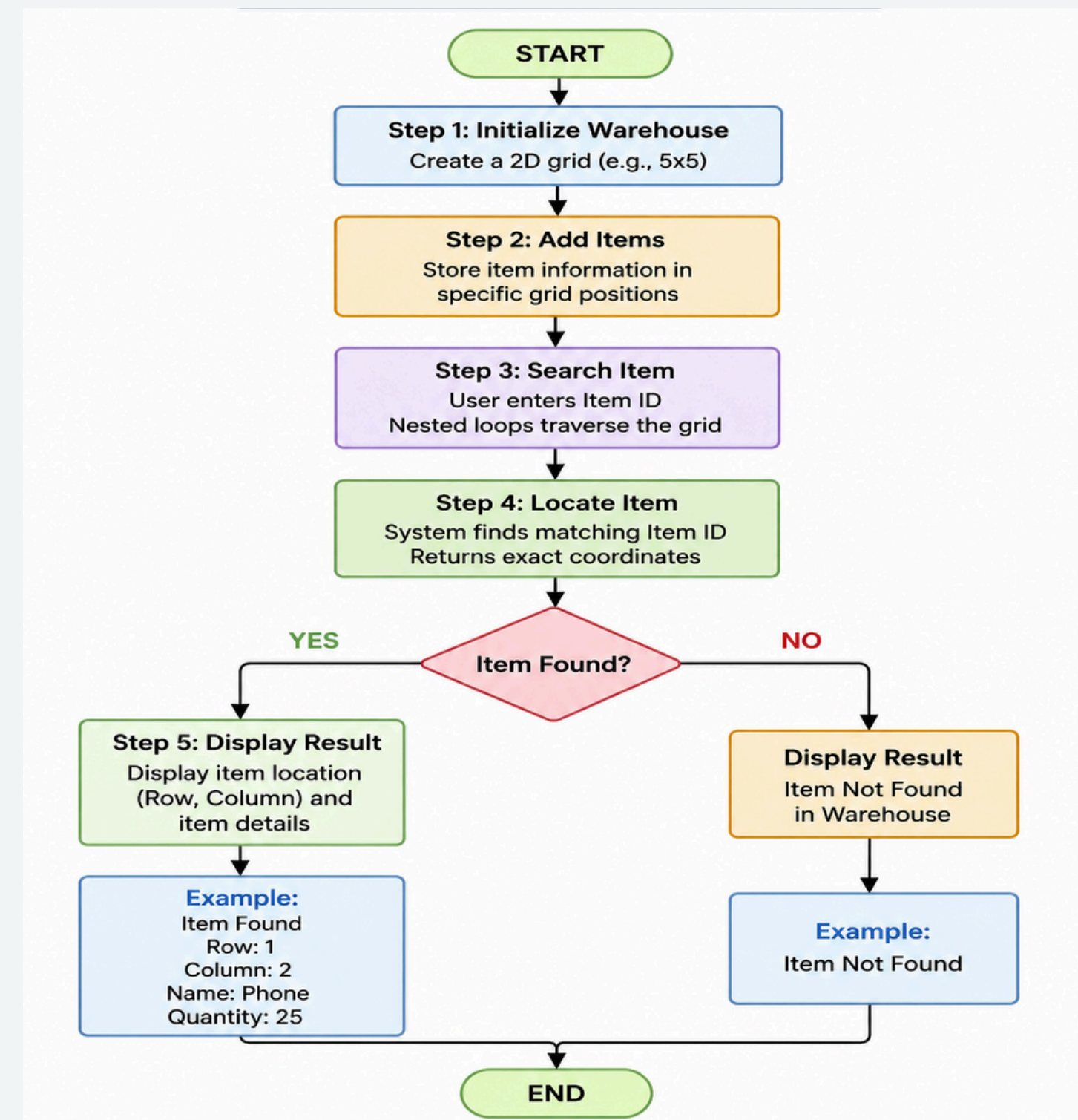
Step 3: Search Item

- User enters Item ID
- Nested loops traverse the grid

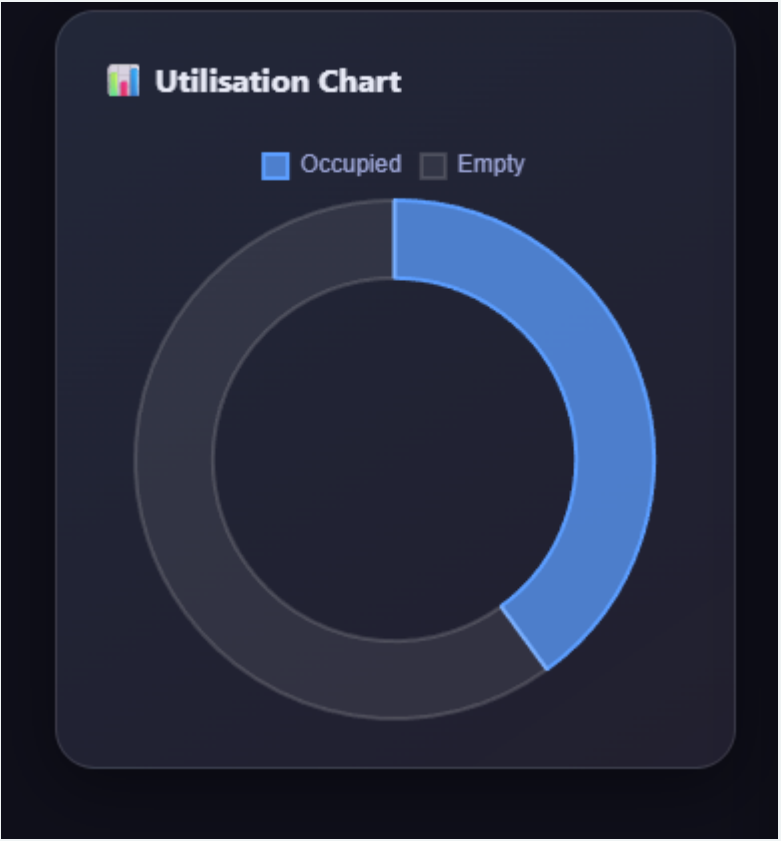
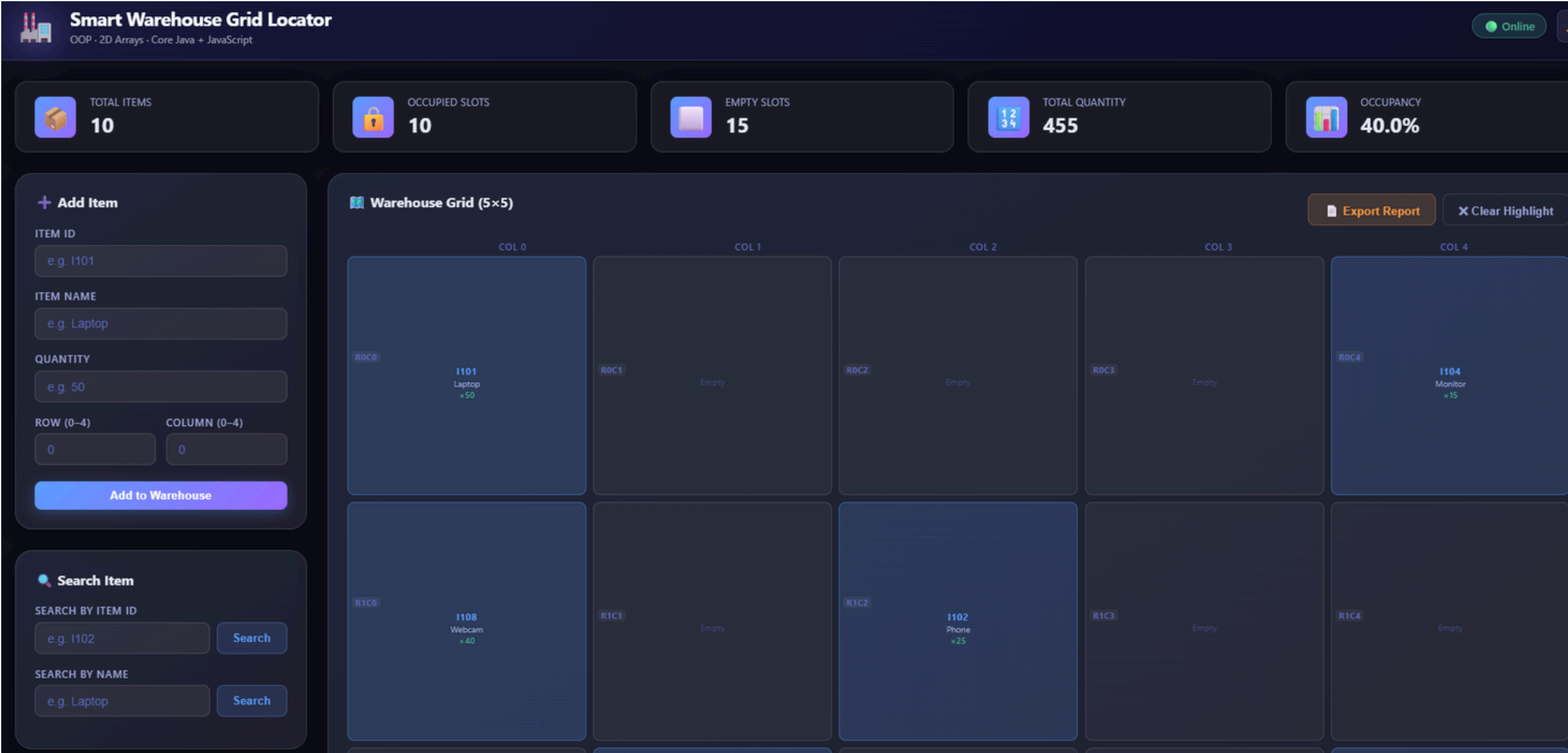
Step 4: Locate Item

- System finds matching Item ID
- Returns exact coordinates

Step 5: Display Result



SCREENSHOTS



SMART WAREHOUSE GRID LOCATOR - REPORT
Generated: 10/6/2026, 3:52:07 pm
Grid Size: 5x5
Occupancy: 40.0%
Occupied: 10 | Empty: 15
Total Quantity: 455

Item ID	Name	Qty	Row	Col
I101	Laptop	50	0	0
I104	Monitor	15	0	4
I108	Webcam	40	1	0
I102	Phone	25	1	2
I103	Tablet	30	2	1
I107	USB Hub	60	2	4
I109	Headphones	20	3	2
I105	Keyboard	100	3	3
I106	Mouse	80	4	1
I110	SSD Drive	35	4	4
TOTAL		455		

Recently Added Search History All Items

ITEM ID	NAME	QUANTITY	ROW	COL	ACTIONS
I101	Laptop	50	0	0	Edit
I104	Monitor	15	0	4	Edit
I108	Webcam	40	1	0	Edit
I102	Phone	25	1	2	Edit
I103	Tablet	30	2	1	Edit
I107	USB Hub	60	2	4	Edit
I109	Headphones	20	3	2	Edit
I105	Keyboard	100	3	3	Edit
I106	Mouse	80	4	1	Edit
I110	SSD Drive	35	4	4	Edit

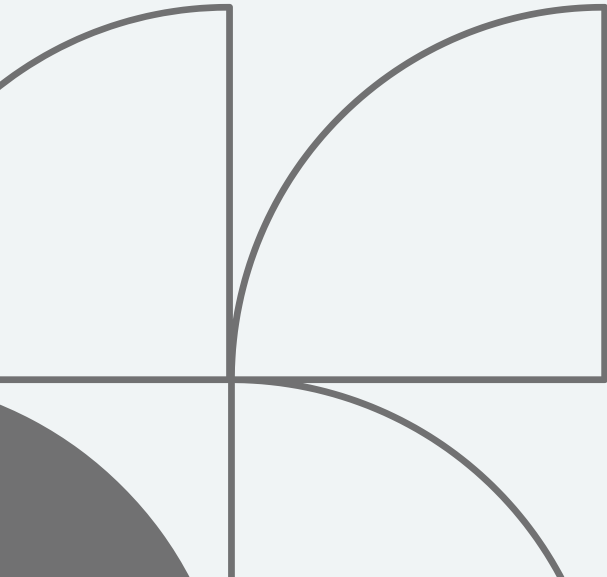
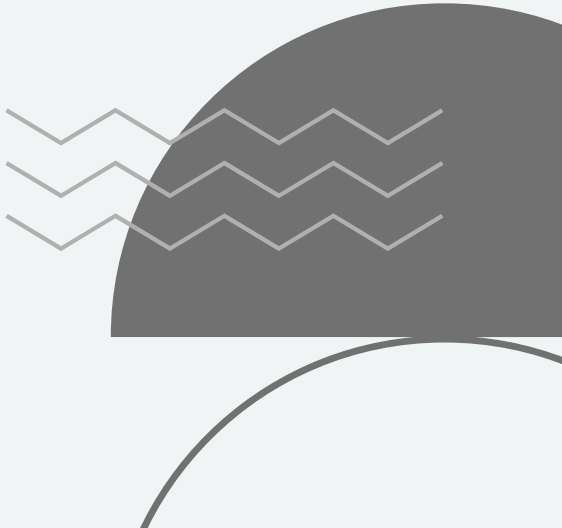
Recently Added Search History All Items

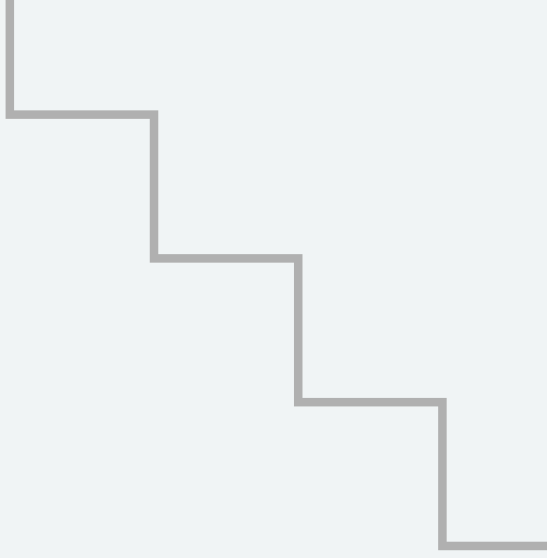
- I110 - SSD Drive | Qty: 35
- I109 - Headphones | Qty: 20
- I108 - Webcam | Qty: 40
- I107 - USB Hub | Qty: 60
- I106 - Mouse | Qty: 80
- I105 - Keyboard | Qty: 100
- I104 - Monitor | Qty: 15



FUTURE SCOPE

FUTURE ENHANCEMENTS

- Barcode Scanner Integration
 - QR Code Tracking
 - Database Connectivity (MySQL)
 - Real-time Inventory Monitoring
 - Cloud-Based Warehouse Management
 - AI-Based Item Recommendation
 - Multi-Warehouse Support
 - Mobile Application Integration
- 
- 



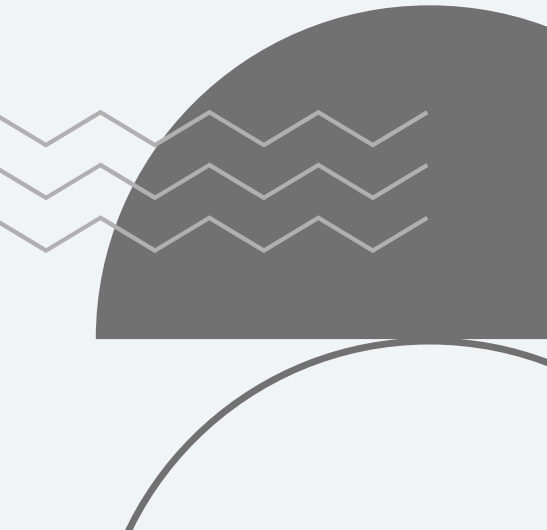
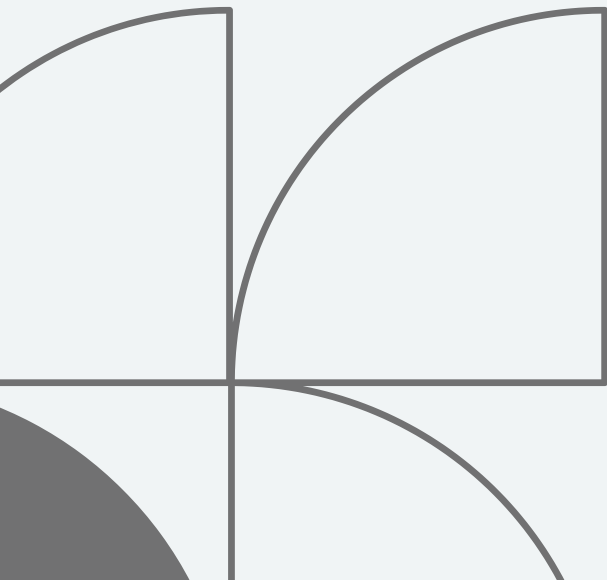
CONCLUSION

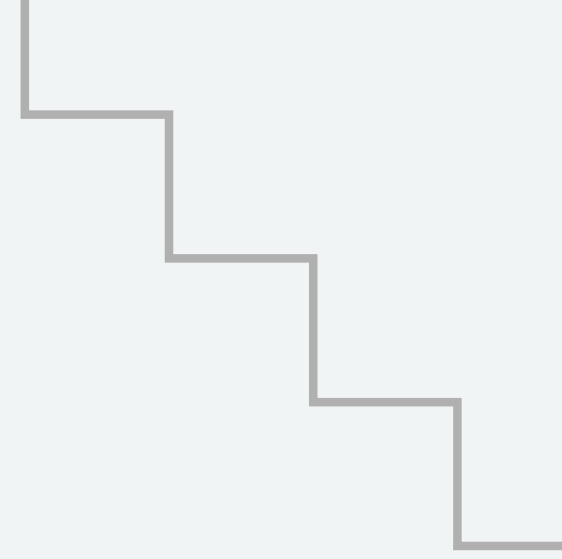
The Smart Warehouse Grid Locator provides an efficient method for storing and locating warehouse items using a 2D grid structure.

LEARNING OUTCOMES:

- Practical use of 2D Arrays
- Object-Oriented Programming implementation
- Search algorithms using nested loops
- Real-world warehouse simulation

The project successfully demonstrates how Java and OOP concepts can be applied to solve inventory tracking problems efficiently.





**THANK
YOU**

